Bradley D. Schmidt

Software Engineering Student

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Summary

Software Engineering student drawn to intricate technical puzzles. Skilled in C++, Python, and JavaScript, with a project background spanning GPU-based ray tracing, mobile apps, and embedded systems. Combines a passion for hands-on exploration with engineering fundamentals from TRU to produce reliable, well-tested solutions.

Education

Thompson Rivers University — Kamloops, BC, Canada
3rd Year Bachelor of Engineering in Software Engineering (GPA: 3.44) — *Expected Graduation Apr. 2027*

Skills

Technical Skills

- **Programming Languages**: C++, Python, RUST, JavaScript, SQL, VHDL, Matlab
- **Developer Tools**: Linux, Git, VS Code, Vim, Android Studio, Oracle SQL Developer
- **AI and ML**: Google Gemini AI, OpenAI API
- **Frameworks**: React, React Native, Node.js
- **Platforms**: Firebase, GitHub
- **Hardware**: Arduino, Raspberry Pi, Esp8266, FPGA

Personal Skills

- **Original Solutions**: Skilled at finding unconventional approaches to complex tasks
- **Methodical Reasoning**: Proficient at breaking down problems into logical, manageable steps
- **Curious Mindset**: Embraces new concepts quickly and enjoys exploring abstract ideas
- **Autonomous Workstyle**: Works independently to craft fresh solutions beyond standard methods

Projects

Ray tracing Engine | Rust, C++ — Feb. 2025 – Present

- Built from scratch using low-level math to calculate ray intersections.
- Modeled light bending around simulated black hole, incorporating gravitational lensing equations directly into ray intersection logic.
- Added diffuse and metallic reflection models, skybox, and adjustable parameters.
- Currently working on implementing Vulkan support and hardware level ray tracing.

AquaFlora Mobile App | *React Native, Google Gemini AI, Python, Realm Database* — Jul. 2023 – Jan. 2025

- Engineered a Mobile app for aquarium management, for users to track water parameters, reminders, and fish stocking.
- Led a 14-day testing phase with 12 users, uncovering and resolving 10+ issues before public launch.
- Integrated Google Gemini AI for fish/plant/disease image recognition, achieving 85% accuracy in trials and optimizing chatbot care tips via Gemini API.
- Automated data collection and validation for 650+ fish species via a Python scraper, reducing manual entry by 95%.
- Set up a monthly subscription model and Google Calendar integration, generating early recurring revenue and improving user retention.

FPGA Calculator | VDHL, Vivado, ModelSim, Basys 3 FPGA — Nov. 2024 – Nov. 2024

- Used Basys 3 FPGA to make a four function calculator.
- Interfaced the FPGA with 4-7 segment display, and PMOD KYPD keypad
- Implemented addition, subtraction, multiplication and division using behavioral modeling in VHDL.
- Collaborated with one partner for CENG 3010 to deliver a fully functional FPGA-based calculator within a 1-week project window

URepair | *C++, React, JavaScript* — Oct. 2024 – Nov. 2024

- Developed a web based platform in a 3-person team, connecting contractors to job opportunities as part of a group project.
- Created backend (C++), and frontend interface (Node.js)
- Implemented optimal algorithms, improving search times by over 60%.

Work Experience

Maintenance — Mar. 2020 – Present
Service Canada — *Kamloops, BC, Canada*

- Maintained building cleanliness during peak COVID lockdown, demonstrating responsibility and attention to detail.
- Performed tasks such as carpet shampooing, pressure washing, and general upkeep to ensure a safe environment.

Concession Worker — Feb. 2023 – Feb. 2024
Senor Froggy — *Kamloops, BC, Canada*

- Supported the main cook in a high paced concession environment during peak hockey event hours.
- Worked efficiently as part of a tight knit team under pressure, enhancing teamwork and time management skills.